

BURHANI H. KATIMBA

COMPUTER ENGINEER | CYBERSECURITY & SOFTWARE DEVELOPMENT

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PROFILE

BEng Computer Engineering graduate (2026) based in Dar es Salaam, with hands-on penetration testing and secure systems engineering experience. Built three portfolio projects spanning a browser-based cybersecurity lab platform, a live classroom quiz system, and a permission-gated local AI agent. Skilled in Python, Django, Linux, Docker Swarm, APIs, and practical security testing. Seeking roles across cybersecurity, backend engineering, and platform engineering.

TECHNICAL SKILLS

Languages: Python, JavaScript, PHP, SQL, Bash

Backend: Django, Django REST Framework, FastAPI, Channels/WebSockets, Tailwind CSS

Security: Penetration testing, Kali Linux, reconnaissance, scanning, enumeration, exploitation, privilege escalation

Infrastructure: Linux, Docker, Docker Swarm, TLS/PKI, mutual TLS, Git/GitHub, AWS fundamentals

Data & AI: PostgreSQL, SQLite, Redis, local LLMs, Ollama, permission-gated tool use

SELECTED PROJECTS

SomaCloud — Cybersecurity Sandbox Platform

Final-year project · Python, Django, Docker Swarm, mTLS

A self-hosted training platform that provisions isolated Kali Linux labs on remote Docker nodes, with browser-based terminals and guided challenges.

- Authored a 59-lab curriculum across nine attack phases, combining theory, walkthroughs, challenge questions, containers, flag submission, and progress tracking.
- Built a resource-aware multi-node scheduler, self-hosted CA with mutual TLS, and zero-touch worker provisioning.
- Implemented encrypted backups and instructor analytics for cohort activity, lab progress, flag review, and PDF export.

ShamsQuiz — Live Classroom Quiz

Python, Django, Channels/WebSockets, Redis, DRF, JWT

A real-time quiz platform where teachers host sessions and students join with a six-character code.

- Implemented WebSocket state sync for a live podium and scoreboard, with time-based scoring and a reusable question bank.
- Documented REST and WebSocket APIs with JWT authentication and CORS policy for mobile clients.
- Added session PDF reports, score charts, and administration tools; deployed with Apache2 and mod_wsgi.

F.R.I.D.A.Y — Permission-Gated Local AI Agent

Python, FastAPI, Ollama

A local AI agent with no direct filesystem access; every tool call passes through an allow, ask, or deny gateway.

- Bound one-time approvals to an exact action and resource to prevent approval reuse for broader operations.
- Hardened file access against path traversal and symlink escapes using resolved-path containment checks.
- Wrote 14 automated tests covering permission boundaries, filesystem policy, and CLI behavior.

EXPERIENCE

Penetration Tester

Aug 2025 – Oct 2025

DIT ICTCoE, Dar es Salaam Institute of Technology

Tested institute systems including Tibasalama and WOSAC across reconnaissance, scanning, exploitation, and privilege escalation; documented findings and remediation recommendations.

Junior Developer · Software Development Practical

Aug 2024 – Oct 2024

Computer Department, Dar es Salaam Institute of Technology

Built and maintained PHP applications on a LAMP/XAMPP stack, covering database design, authentication, and user interface implementation.

EDUCATION

Bachelor of Engineering in Computer Engineering (BEng-COE)

Oct 2022 – Sep 2026

Dar es Salaam Institute of Technology

Final-year project: Development of a Cybersecurity Sandbox Environment, delivered as SomaCloud.

CERTIFICATIONS & TRAINING

- Generative AI & AI Basics — Vodacom Digital Skills Hub with Microsoft
- AMD AI Developer Program — in progress
- AWS Educate — cloud computing foundations
- TryHackMe — active member; completed courses and CTF challenges

LANGUAGES

Swahili (native) · English (fluent) · Arabic (intermediate) · Turkish (beginner)

REFERENCES

Available upon request.